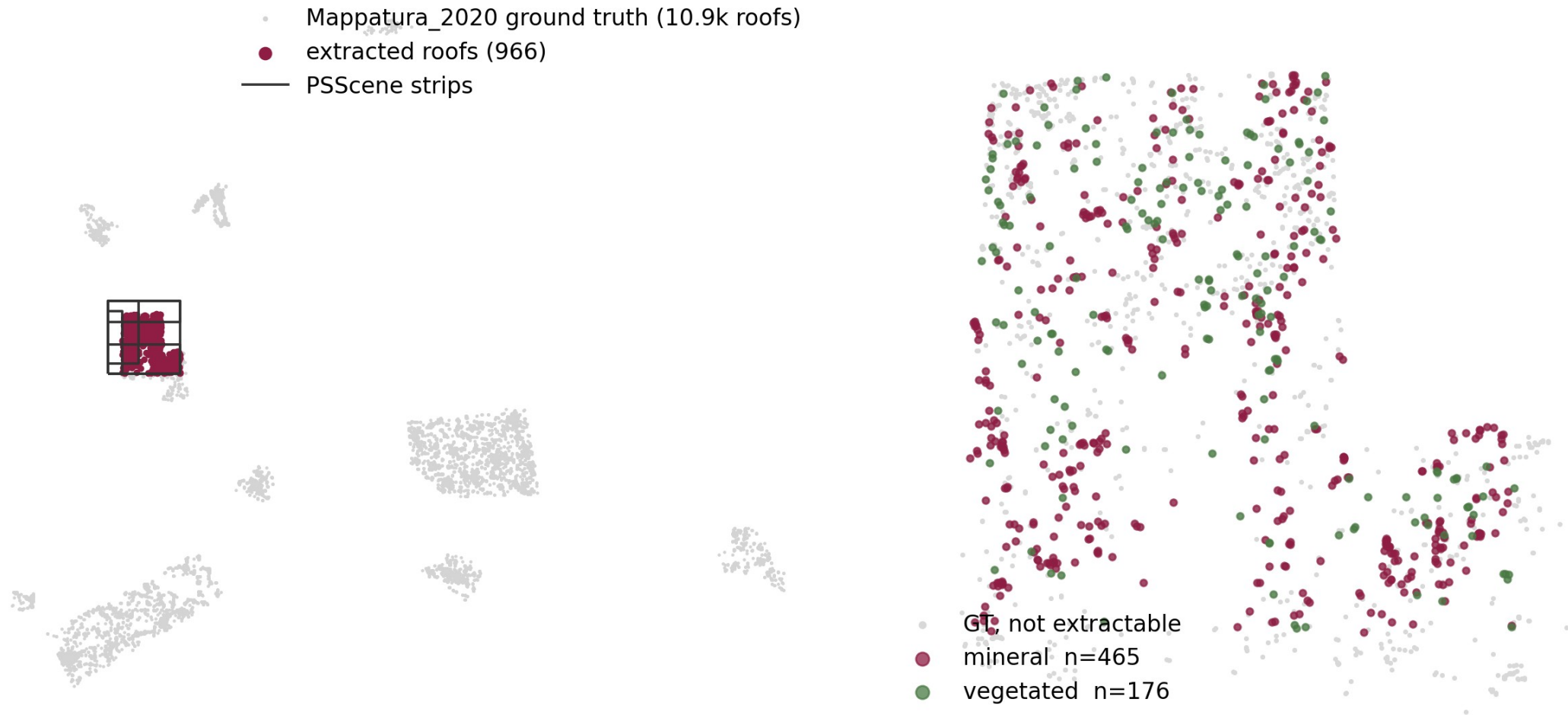


Asbestos roofs: first real spectral signatures from SuperDove imagery

Regional asbestos ground truth (Mappatura 2020) × Planet PSScene 8-band, 12 May 2026

Alessandro Potenza — 12 June 2026

The PSScene strips cover 1,096 mapped roofs — 966 usable signatures



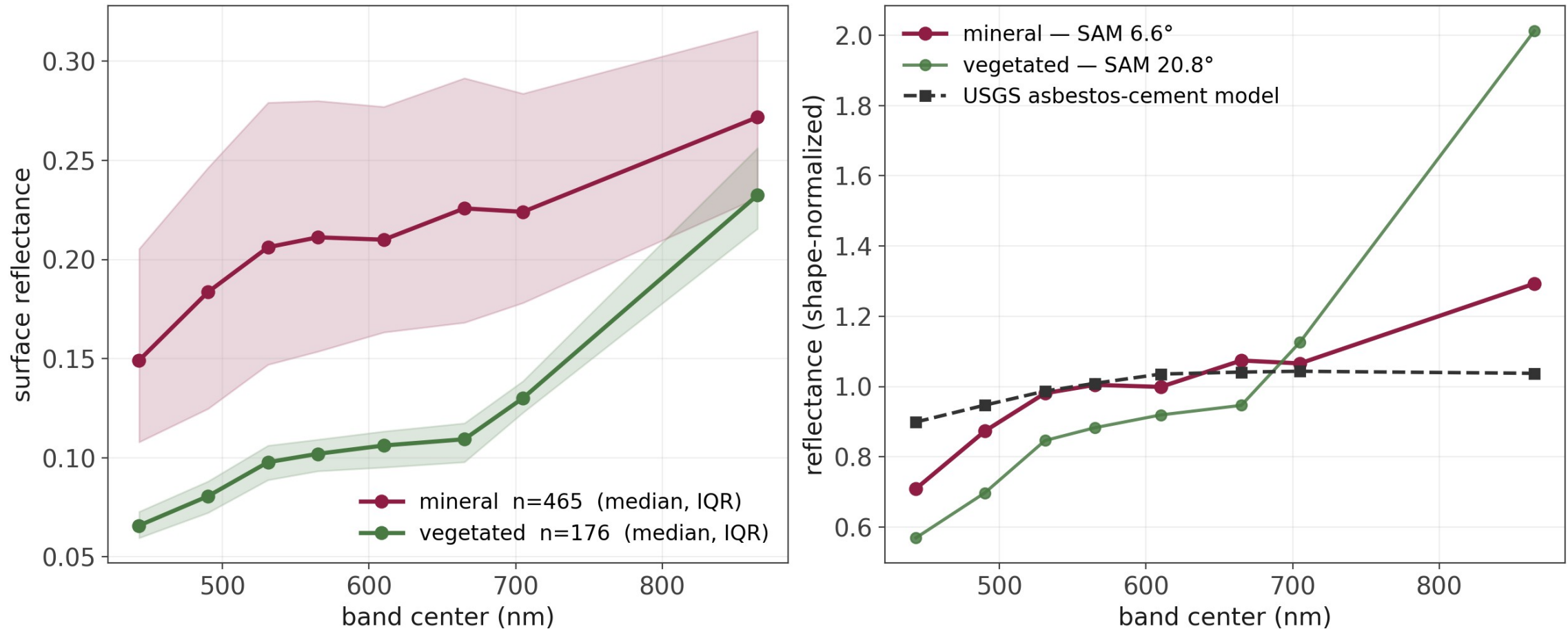
Extraction pipeline: from GT polygons to one signature per roof

- 1 GT selection** Mappatura 2020 census (most recent regional mapping), area $\geq 150 \text{ m}^2$
- 2 Edge erosion** polygon buffered -3 m inward: at 3 m GSD this removes mixed roof/ground border pixels, keeping pure roof pixels only
- 3 Pixel quality** Planet udm2 clear mask; a roof needs ≥ 5 valid pixels
- 4 Signature** per-band median of surface reflectance — 8 SuperDove bands (Coastal Blue $\rightarrow$ NIR); overlapping strips $\rightarrow$ keep the observation with more clear pixels
- 5 Vegetation screening** $\text{NDVI} < 0.2$ mineral (bare roof) · $\text{NDVI} \geq 0.3$ vegetated (likely remediated or overgrown since the 2020 mapping)
- 6 Similarity** SAM — angle between signatures as vectors: compares spectral shape, ignores overall brightness

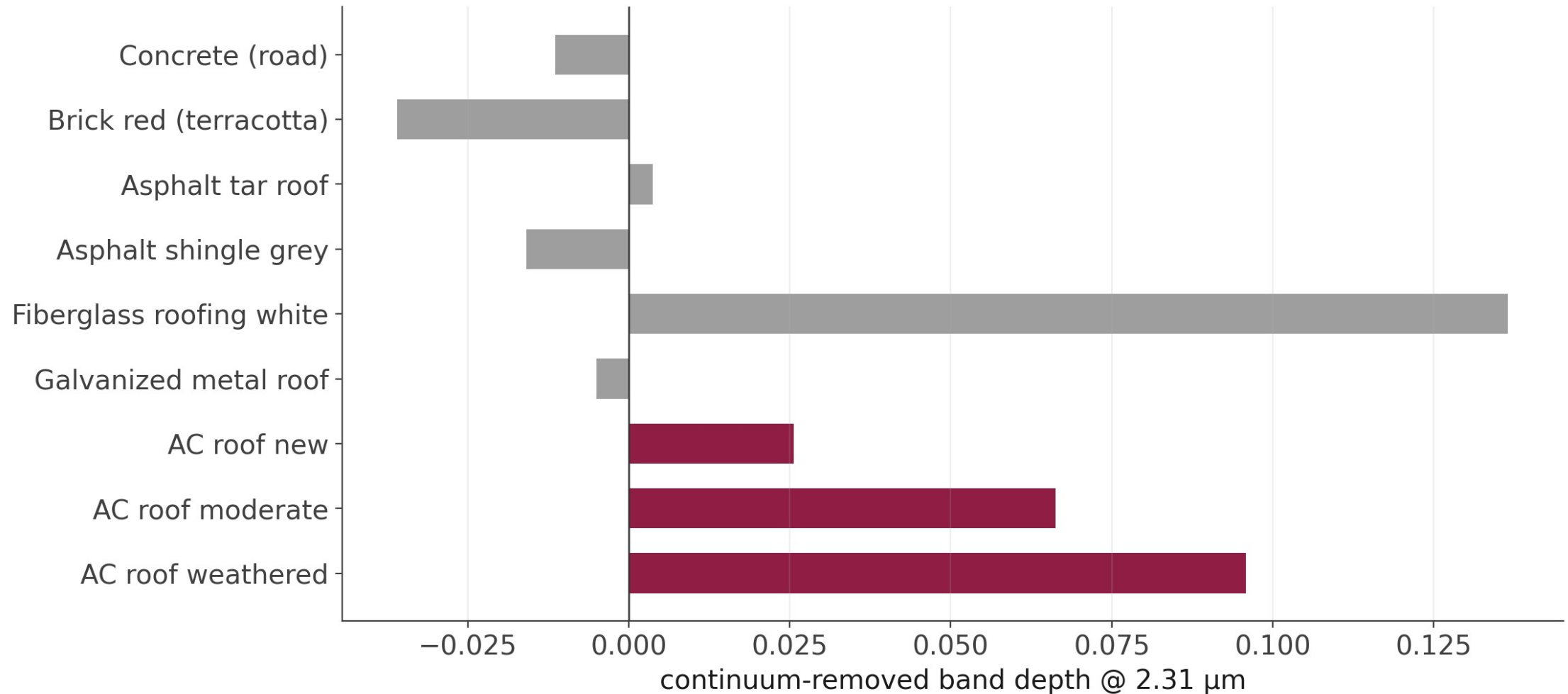
Extraction check on the imagery: GT outlines match the buildings



Bare roofs match the lab reference — NDVI flags likely-remediated roofs



Why SWIR matters: only asbestos-cement shows the 2.31 μm Mg-OH feature



What the VNIR bands can and cannot do

Asbestos-cement vs terracotta: separable

SAM 21.6° on SuperDove bands — the common Lombardy roof is not a confuser

Asbestos-cement vs bare concrete: not separable

SAM ~4° on every band set — the hard confuser stays hard in VNIR

The diagnostic feature lives in the SWIR

Mg-OH absorption at 2.31 μm : needs WV-3; out of reach for SuperDove and Sentinel-2

Next step

extend the extraction to more covered roofs and validate the SWIR feature on the project WV-3 scenes